University of Nottingham
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Department: [Electrical and Electronic Engineering]

WHEN IS 3D WORTH IT?

Task-Conditioned Adequacy of Planar Localisation for Vision-Guided Manipulation

OUR CONTRIBUTION We combine analytically predicted off-plane error with a separate real-system error bound to determine whether planar localisation meets a specified task tolerance.

VALIDATION SETUP & ROBOT CONTEXT



RoArm M2
Robot context only



Berxel iHawk 100Q
E2 physical validation
2 × RGB-D + marker board

1. INTRODUCTION

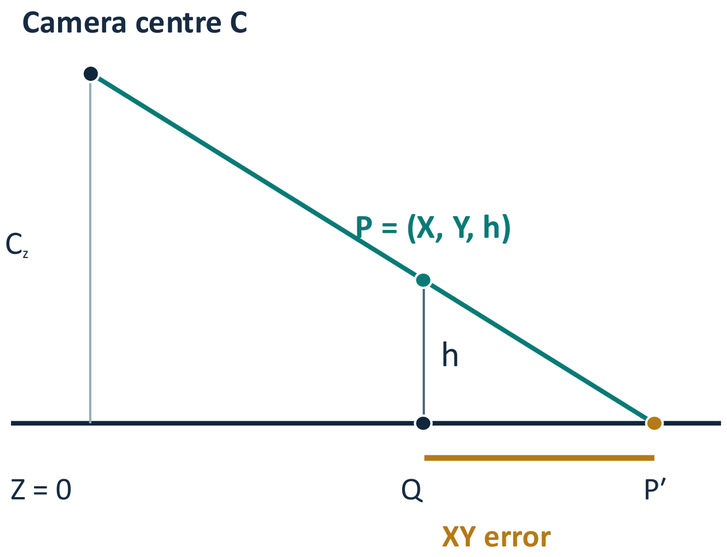
When is 2D localisation good enough?

Planar localisation is simple and efficient. Off-plane targets introduce systematic XY positioning errors. [1, 2]

Can we predict these errors, and when do they require 3D-aware localisation for a task guarantee?

Aim: predict, validate and assess task sufficiency.

Figure 1 Off-plane localisation



Q: true lateral target projection. P′: planar estimate. Z: vertical coordinate; Z = 0: calibration plane. The viewing ray passes through C, P and P′.

2. METHODOLOGY

2.1 Analytical geometric model

$$\Delta X = h(X - C_x) / (C_z - h)$$

X-direction bias: planar estimate minus true X.

$$\Delta Y = h(Y - C_y) / (C_z - h)$$

Y-direction bias: planar estimate minus true Y.

$$\rho = \sqrt{[(X - C_x)^2 + (Y - C_y)^2]}$$

Lateral camera–target distance in the XY plane.

$$E_{geom} = |h| \rho / |C_z - h|$$

Magnitude of the geometry-induced XY error.

C = (C_x, C_y, C_z): camera centre (XY, height).
P = (X, Y, h): target; X, Y are work-plane coordinates.
h: target height; C_z: camera height above Z = 0.
All lengths: mm; $|\cdot|$: absolute value; √: square root.
Board-relative h; fixed rotation; robot Z unverified.

2.2 Blind physical validation

E2: two RGB-D cameras + calibrated marker board.
5 rebuilds; 5 locations; h = 0:5:50 mm.
Z = 0 fitting; no off-plane fitting or grasp-success test.

Calibrate at Z = 0 › Predict geometry › Measure off-plane › Compare errors

2.3 Task-based bounded-error analysis

$$e_P = e_{geom} + r_P$$

e_P: total XY; r_P includes baseline mapping error.

$$\|r_P\| \le S_P$$

S_P: separately declared bound; not residual RMSE.

3. RESULTS

3.1 Analytical geometric error — E2 cameras

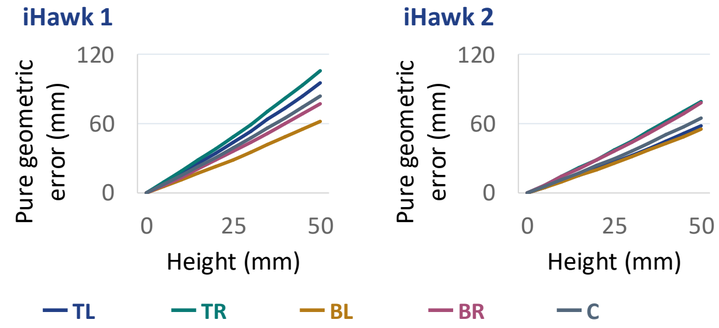
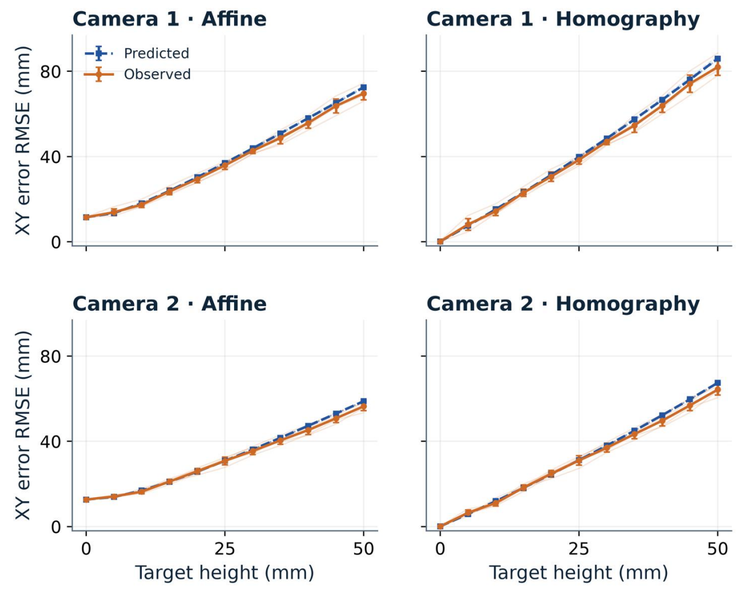


Figure 2. Analytical prediction using reconstructed E2 camera geometry. Each marker: mean of 5 rebuilds; board-frame XY and height (mm). C_z: iHawk 1, 356.13–359.22; iHawk 2, 349.46–351.10 mm.

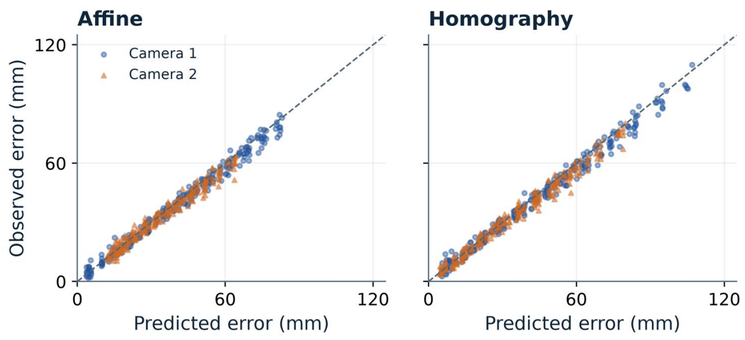
3.2 Blind experimental validation

3A Blind Geometric Prediction vs Observation



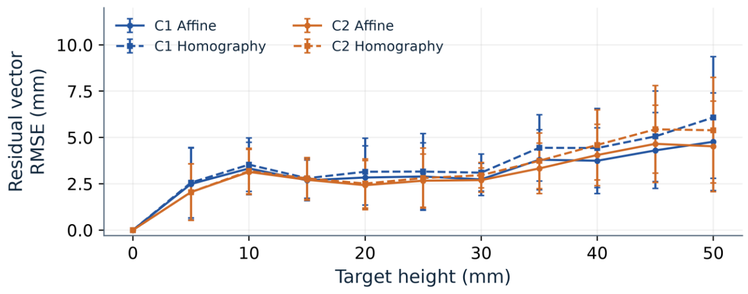
$e_{pred,E2} = e_{fit,0} + \delta_M(h)$ (vector sum)
$e_{fit,0}$: Z = 0 fit error; δ_M: frozen-mapper increment.
Mean ± SD: 5 rebuilds; faint lines: individual rebuilds.
Height-curve r = 0.9942–0.9951 (h > 0); no h > 0 fitting.

3B Point-Level Prediction Agreement



5–50 mm; 250 points per camera/model; n = 5 rebuilds.
Magnitude r = 0.9869–0.9942; median direction cosine = 0.9983–0.9992. Correlation is not absolute accuracy.

3.3 Geometry-Unexplained Residual Analysis



$r_{E2} = e_{obs} - e_{pred,E2}$ (not task remainder r_P).
Residual RMSE: 3.61–4.36 mm; SD bars: 5 rebuilds.
Remaining error requires a separate bound; RMSE ≠ S_P.

3. RESULTS CONTINUED

3.4 STAR-loc external validation summary

12,473 repeated observations
55 AprilTags; 8 held-out sequences

0.981	**0.998**
Spearman magnitude correlation	Median direction cosine

Independent observations support the structural projective-error mechanism [A, §7.3; 3]. Supporting evidence only: no absolute-accuracy or task-boundary validation.

4. DISCUSSION

When is 3D actually required?

01 $e_P = e_{geom} + r_P$
r_P covers all error beyond geometry, including baseline.

02 $\|r_P\| \le S_P$
S_P: declared upper bound on remaining XY error.

03 $E_P = \|e_P\| \le E_{geom} + S_P$
E_P: total error magnitude; $\|\cdot\|$: Euclidean norm.

04 $E_{geom} + S_P \le \tau$
τ: allowable task XY error (mm); planar use is certified.

05 $E_{geom} + S_P > \tau$
Planar guarantee unavailable; actual failure is not implied.

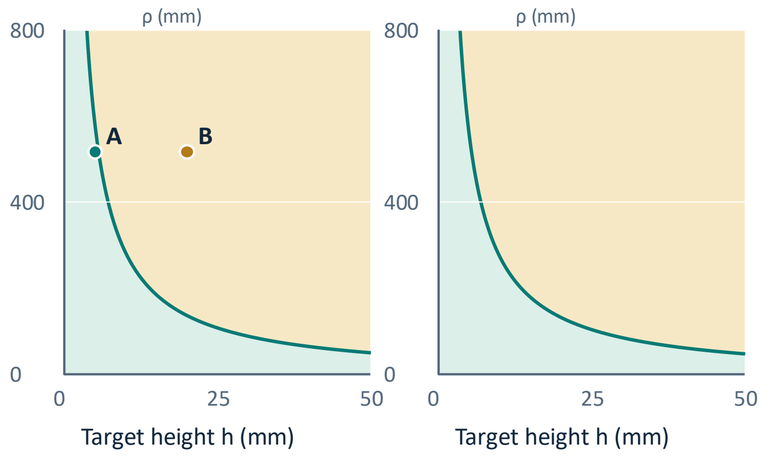
06 $S_{3D} \le \tau$
S_{3D}: declared 3D bound; with row 5 → 3D-only guarantee.

Figure 6 Task-Conditioned Adequacy

E2 Camera Geometry with Illustrative Error-Bound Scenarios

E2 geometry: median-C_z rebuild 20260808_162715.
τ = 10, S_P = 2, S_{3D} = 5 mm: hypothetical scenarios.
Not experimentally certified; S_P is not an E2 error bound.

Teal: both certified Gold: only 3D certified



Boundary: $h\rho/(C_z - h) + S_P = \tau$; $0 \le h < C_z$.

iHawk 1; E2 centre marker (X, Y) = (0, 0) mm

A h = 5 mm; ρ = 516.47 mm
E_{geom} = 7.29 mm
Planar bound: 9.29 mm < τ
BOTH GUARANTEED

B h = 20 mm; ρ = 516.47 mm
E_{geom} = 30.47 mm
Planar bound: 32.47 mm > τ
3D ONLY GUARANTEED

"Not guaranteed" does not mean actual failure. Camera geometry is E2-derived; task tolerance and error bounds are illustrative, not certified.

5. CONCLUSIONS

Geometry matters

$E_{geom} = |h| \rho / |C_z - h|$

Height alone cannot determine localisation adequacy.

Blind prediction validated

0.9942–0.9951

Predictions capture the dominant experimental error trends.

Real-system errors remain

3.61–4.36 mm

Geometry alone does not fully explain total localisation error.

Task tolerance determines adequacy

$E_{geom} + S_P \le \tau$

Planar localisation can remain sufficient under the declared error bound.

3D IS NOT ALWAYS NECESSARY, EVEN WHEN 2D IS GEOMETRICALLY IMPERFECT.

[1] J. Carlson, "Flatland Revisited: The Effects of Non-Planar Floors on Large-Scale Warehouse Deployment of Autonomous Mobile Robots," ICRA Robots in the Wild Workshop, 2025.
[2] J. Carlson & A. Kolling, "Selective Placement of Visual Fiducials for Localization Tasks," ICRA Robot Execution Failuresand Failure Management Strategies Workshop, 2023.
[3] F. Dümbgen et al., "STAR-loc: Dataset for STereo And Range-based localization," arXiv:2309.05518, 2023. [A] Research drafts. E2 frozen package: final-SEP, commit 9a2844f.

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